**Initial Plan: Advanced Music Audio Alignment Using Differentiable Dynamic Time Warping and Deep Learning Representations**
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## Project Description

The current global music industry is experiencing a "sea of noise" because of the democratisation of production tools and generative AI [1, 2]. This means there has never been so much content in the history of music. It is now more important than ever to automatically and accurately synchronize the audio recordings with symbolic representations of the recordings (scores, lyrics, or MIDI) [3]. This is very important for the application of automatic score following, remixing and the deep musicological analysis of recordings.

## Core Problem This Project Addresses

The main issue this project aims to solve is the limitations of the classical Dynamic Time Warping (DTW) for the alignment of musical audio recordings. Although DTW is a widely used method for the alignment of temporal sequences, it generally depends on manually defined features (e.g., Chroma vectors) that cannot change before alignment. Furthermore, the classical DTW operation includes a discrete minimum function that is not differentiable. Because of this property, DTW can't be used as a loss function in deep neural networks that would allow us to train the network to find optimal feature representations "end-to-end" from raw audio [4, 5].

## Implementation of Differentiable Dynamic Time Warping (Soft-DTW)

In order to close this gap, I will implement Differentiable Dynamic Time Warping (Soft-DTW) [6] inside a deep learning framework. Unlike classical DTW, Soft-DTW replaces the hard minimum with a smoothed operator, allowing gradients to back-propagate through the alignment process. This enables end-to-end training of a Convolutional Recurrent Neural Network (CRNN) that learns an embedding space **specifically optimised for aligning two audio recordings of the same musical work**. Concretely, the network will map each recording into a time-indexed feature sequence, and Soft-DTW will be used as the differentiable alignment loss between the two sequences, encouraging the learned representation to make corresponding musical events similar while remaining robust to tempo variation.

## Scope of Project

The main aim of this project is the development and evaluation of the DeepAlign-26 system for **audio-to-audio alignment**, i.e., given **two different recordings/performances of the same piece**, estimate the temporal warp that best aligns them. The system will be trained offline and evaluated against established baselines, using Soft-DTW to align the learned feature sequences extracted from each recording. Where datasets provide score-derived or annotation-based timing information (e.g., SWD [8] and MazurkaBL [9]), these will be used to construct training/evaluation targets and to quantify alignment error, but the **core alignment task and output remain recording-to-recording correspondence**. As a secondary goal, I will investigate whether the learned embeddings also transfer to **online** alignment settings (e.g., real-time following) using Matchmaker [7], by replacing hand-crafted features in the pipeline with the learned representation.

**Minimum Deliverable**

At least, this project will produce a robust offline alignment system that is trained on the SWD, compared with Synctoolbox [10] and Matchmaker baselines, and that has reproducibly generated code and results tables.

**Aims and Objectives**

**Overall objective:** To develop and evaluate a state-of-the-art music alignment system, which outperforms the existing classical methods by learning optimal feature representations via end-to-end trainable alignment loss functions.

**Objective 1: Developing a Deep Learning Architecture that Implements the End-To-End**

**Trainable Differentiable Alignment Loss Function**

- **Goal:** Develop a PyTorch based CRNN architecture that uses CUDA accelerated Soft-DTW to learn alignment features directly from pairs of audio recordings of the same musical work.
- **Risk:** Large computational costs arising from the $O(N^2)$ complexity of DTW during training.
- **Mitigation:** Use efficient CUDA implementations (e.g., pytorch-softdtw-cuda) and implement "Weakly Aligned" training strategies to shorten the sequences during the initial coarse-training phase.

**Objective 2: Comparison with Established Industry Standards**

- **Goal:** Compare the deep learning model with Matchmaker (Online Time Warping) and Synctoolbox (Offline DTW) on the SWD test data.
- **Evaluation Protocol:** The performance will be determined by calculating the mean absolute error (MAE) between estimated alignment correspondences and ground-truth correspondence annotations using a 10 ms frame hop size. An estimated "match" is counted if the alignment error is within a specified tolerance window ($\theta$).
- **Success Criteria:**
    - **Success Criterion 1: Robust Alignment.** Achieve alignment performance comparable to established hand-crafted Chroma features (MAE < 50ms) on the

SWD dataset.
    - **Success Criterion 2: State-of-the-Art Performance.** Achieve State-of-the-Art (SOTA) performance (MAE < 20ms and AR > 98%) at a tolerance window θ = 50ms.
- **Risk:** Overfitting to the acoustic properties of the training data.
- **Mitigation:** Use data augmentation techniques (random time-stretching ±20%, random pitch shifting, random additive noise) to enforce generalized feature learning.

**Objective 3: Robustness Evaluation using Datasets with Extreme Rubato**

- **Goal:** Test the developed system on the MazurkaBL dataset [9], analyzing the alignment performance under conditions of extreme rubato.
- **Risk:** Human annotation errors or inconsistencies in the ground truth.
- **Mitigation:** Use automatic monotonicity checkers and outlier detectors during data preprocessing; utilize the Soft-DTW loss that is robust to "weak" (coarse) labels.

## Feasibility

**Computational Resources:**

As the project requires GPU acceleration to train deep neural networks and compute Soft-DTW loss, I will rely on the GPU capabilities of my workstation. My workstation is equipped with a CUDA enabled GPU that can handle the required PyTorch/CUDA workload.

- **Fallback strategy:** If I encounter issues with the CUDA implementation, I will resort to a CPU-based Soft-DTW implementation that utilizes shorter sequence lengths and gradient accumulation to manage training times.

**Data Availability and License:**

I have acquired access to all necessary datasets for research purposes. I have acquired access to both the Schubert Winterreise Dataset (SWD) and the MazurkaBL dataset.

- **Compliance:** I will follow the Creative Commons Attribution 3.0 license of the SWD and the CC BY-NC-SA 4.0 license of the MazurkaBL dataset. I will not redistribute any original audio data; only experimental results and derived feature descriptors will be published.

**Required Software Packages:**

All core dependencies (PyTorch, partitura, madmom, Matchmaker, Synctoolbox) are open-source Python packages. Since my project utilizes publicly available datasets and involves no human subjects, I do not need any ethics approval.

## Work Plan

**Phase 1: Foundations and Baselines (Weeks 1–3)**

- **Week 1 (February 2 – 8 February):** Environment setup: configure Python environment (PyTorch, CUDA); verify compilation of Soft-DTW CUDA kernels; meeting with supervisor to agree on schedule.
- **Week 2 (February 9 – 15 February):** Data preprocessing: write scripts to transform SWD audio to Constant-Q Transform (CQT) spectrograms (frame hop size: 10 ms) and prepare pairing/metadata to form matched recording pairs; generate weakly aligned training targets using available annotations (and optionally score-derived features

where available).

- **Week 3 (February 16 – 22 February):** Baseline evaluation: run standard Matchmaker and Synctoolbox algorithms on the SWD test data to determine the MAE/AR floor; milestone: baseline performance report.

**Phase 2: Core Work (Weeks 4–7)**

- **Week 4 (February 23 – March 1):** Model implementation: develop the dual stream CRNN encoder and custom SoftDTWLoss module in PyTorch.
- **Week 5 (March 2 – 8 March):** Training loop: begin full training on SWD; tune smoothing parameter γ (annealing from 1.0 to 0.01).
- **Week 6 (March 9 – 15 March):** Augmentation pipeline: implement on-the-fly time-stretching and pitch-shifting to avoid overfitting.
- **Week 7 (March 16 – 22 March):** Spring break/buffer: refine architecture (possible addition of attention layers); continue training; milestone: trained DeepAlign model.

**Phase 3: Evaluation and Analysis (Weeks 8–10)**

- **Week 8 (March 23 – 29 March):** Stress testing: test the trained model on the unseen MazurkaBL dataset; analyze the warping paths for handling of rubato.
- **Week 9 (March 30 – April 5):** Comparative analysis: prepare final metric tables comparing DeepAlign with Matchmaker and Synctoolbox.
- **Week 10 (April 6 – 12 April):** Statistical validation: perform Friedman/Nemenyi significance tests on the results; create visualizations (warping paths, error histograms); milestone: final results tables.

**Phase 4: Writing and Submission (Weeks 11–12)**

- **Week 11 (April 13 – 19 April):** Report draft: synthesize findings; link technical results with the "Sea of Noise" problem statement.
- **Week 12 (April 20 – 26 April):** Final polish: clean up code documentation; proofread final report; submit deliverables: Final PDF Report, GitHub Code Repository, Trained Model Weights, and Reproducibility Instructions.

**References**

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